

(58 PTS total)

References: Lecture Handouts - *Chart of Nuclides Key, Co-60 Activation Levels, Uranium Decay Series, Table 3-11 Tabulated Reactor Data, Properties of Elements - Lamarsh Table II.3; Select U-Np-Pu Isotopes.*

Read: Reactor Theory Chapter #1: §1.1 – §1.3

1. **6 PTS** - What are the following values for Cobalt (*see key and ^{60}Co handout*)?
 - a) The ^{59}Co resonance integral leading to the Isomeric State? 39b
 - b) The ^{59}Co thermal neutron capture cross section leading to the Ground State? 16b
 - c) The half-life of the ^{60}Co Ground state = 5.271a
 - d) The half-life of the ^{60}Co Metastable (Isomeric) state = 10.47m
 - e) The decay modes of the ^{60}Co Ground state = β^- (0.318 MeV); γ (1332.5, 1173.2 keV)
 - f) The decay modes of the ^{60}Co Metastable (Isomeric) state = IT 58.6; β^- (1.6), γ (1332.5)
2. **1 PT** - Give an example of an isotope that is naturally occurring and radioactive: ^{238}U , $^{234/235}\text{U}$, ^{138}La , ^{218}Po (accept other examples)
3. **1 PT** - Does the cross section (i.e. probability) of undergoing an absorption reaction (σ_a) generally increase or decrease as a fast neutron moderates and loses energy? increases
4. **2 PTS** - What is the decay chain leading from ^{234}U to ^{222}Rn ? $^{234}\text{U} \rightarrow ^{230}\text{Th} \rightarrow ^{226}\text{Ra} \rightarrow ^{222}\text{Rn}$
5. **4 PTS** - What are the two types of neutron absorption reactions? Give an example of each as they might occur inside a reactor core.
 - **Capture** – e.g. $^{27}\text{Al}(n,\gamma)^{28}\text{Al}$, $^{238}\text{U}(n,\gamma)^{239}\text{U}$, etc...
 - **Fission** – e.g. $^{235}\text{U}(n, \text{fission})$; $^{239}\text{Pu}(n, \text{fission})$
6. **2 PTS** - What is the difference between elastic and in-elastic neutron scattering?
 - **For elastic scattering, kinetic energy is conserved. For in-elastic scattering, some kinetic energy is lost to the excitation of the scattered nucleus.**
7. **3 PTS** - Give examples of:
 - a fertile nuclide: ^{238}U , ^{234}U , ^{232}Th
 - a fissile nuclide: ^{233}U , ^{235}U , ^{239}Pu
 - a fissionable nuclide: ^{238}U , $^{232}\text{Th} + ^{233}\text{U}$, ^{235}U , ^{239}Pu
8. **2 PTS** - What is the total amount of energy released from the fission of a ^{235}U nucleus? Is more energy given off immediately (promptly) or delayed?

~210 MeV total energy is released per fission of a ^{235}U nucleus. ~185 MeV is given off promptly, with ~25 MeV coming off delayed.

9. **10 PTS (0.5 ea)** - Reference Lamarsh Table II.3 – ‘Properties of the Elements and Certain Molecule’ and fill in the values in the tables below:

Which material has the best moderating properties?:

Material	Σ_a (cm ⁻¹)	Σ_s (cm ⁻¹)	Ratio: Σ_s/Σ_a
(Light) Water (H ₂ O)	0.0222	3.443	155
Heavy Water (D ₂ O)	0.00004420	0.4519	10224
Beryllium	0.001137	0.7589	667
Graphite	0.0002728	0.3811	1397

Which material would make the best control rod absorber (based on these parameters alone)?

Material	Atom Density (N_a) (atoms/cm ³) $\times 10^{24}$	Σ_a (cm ⁻¹)
Boron	0.1281	97.23
Cadmium	0.04635	113.56
Gadolinium	0.03045	1492
Lead	.03296	0.005603

See lab questions – next page!

Lab Questions – PULSTAR Systems and Schematics**REF: Lab Handouts & Schematics**

10. **9 PTS** - What are the following values and parameters for a 25 Assembly PULSTAR Core?

- Fuel material & % Enrichment: UO₂, 4% ²³⁵U
- Fuel Cladding material: Zircaloy-2
- 5X5 Core Fuel mass = 359 kg UO₂
- Mass ²³⁵U in fresh core = 12.59 kg +- 0.5kg
- Length of Active Fuel = 24 inches
- Moderator material: Light Water / H₂O
- Reflector material(s): Graphite, Beryllium, Water (Accept any of these)
- Control Rod Absorber material(s): Silver, Indium, Cadmium
- Control Rod Clad material(s): Tin & Nickel

11. **8 PTS** - SCRAM Matching – match the SCRAM to the setpoint (*answers may be used more than once, or not at all...*)

<u>Ans.</u>	<u>SCRAM</u>	<u>Setpoint</u>
<u>H</u>	1. Linear Power Channel – Over Power SCRAM	A. 116°F
<u>H</u>	2. Safety Power Channel – Over Power SCRAM	B. 150 kW
<u>B</u>	3. Log-N Channel – Flow/Flapper SCRAM Enable	C. 2000°F
<u>B</u>	4. Safety Power Channel – Flow/Flapper SCRAM Enable	D. -36 inches
<u>A</u>	5. T ₂ Pool Temp	F. 475 GPM
<u>D</u>	6. Pool Level	G. Red Button
<u>F</u>	7. Primary Coolant Flow	H. 1.20 MW
<u>G</u>	8. Manual SCRAM	J. 2.05 MW

12. **2 PTS** - What are the trips associated with the Source Range Channel? < 2CPS; >9E4 CPS

13. **2 PTS** - Does the PULSTAR have a confinement or a containment? Confinement
 What is its main purpose? To contain/confine airborne radioactive material inside the reactor bay utilizing negative pressure with a filtered exhaust flow.

14. **3 PTS** - What are the three (3) main functions of the primary water?

Coolant, moderator, shielding

15. **1 PTS** - What is the function of the three delay tanks in the PPV?

To delay primary flow to allow time (~ 3min) for ¹⁶N to decay away.

16. **2 PTS** - Which describes the correct flow path of heat from the reactor core to the atmosphere?

- Core, Primary Pump, ¹⁶N Delay Tank, Heat Exchanger, Cooling Tower
- Core, Heat Exchanger, ¹⁶N Delay Tank, Primary Pump, Cooling Tower
- Core, ¹⁶N Delay Tank, Heat Exchanger, Primary Pump, Cooling Tower
- d. Core, ¹⁶N Delay Tank, Primary Pump, Heat Exchanger, Cooling Tower**